TsungWei Cheng Mike

Academic Curriculum Vitae

TsungWei Cheng (Mike) is a researcher specialising in digital fabrication, timber–earthen prototyping, and modular architectural systems. He holds a master's degree from NTUST and has completed academic exchanges at TU Berlin and TU Darmstadt.

He is currently a Research Collaborator at IBOIS–EPFL, developing bio-sourced architectural components that bridge low-tech materials with high-tech fabrication. He has published over ten papers on digital fabrication, architectural education, and modular systems.

His work focuses on transforming vernacular construction with bio-sourced materials into industrialised, digitally fabricated building methods, aiming to expand their architectural applications while reducing carbon emissions.

Professional Experience

Research Collaborator
Lausanne, Switzerland July 2026 - Present
Research Team: IBOIS, Prof. Yves Weinand
École Polytechnique Fédérale de Lausanne
Research focus: Bio-source materials and prototypes

Research Assistant
Taipei, Taiwan, Apr 2023 - June 2025
Research Team: GAS Lab, Prof. ShenGuan Shih
National Taiwan University of Technology and Science
Research focus: Dry Stacking System with Hierarchical Assembling by Rule-Based Approach and Interlocking Blocks Modular Design

Lecturer
Kaohsiung, Taiwan, Nov 2024 - June 2025
National University of Kaohsiung
Lecture Topics: Serious Gaming, innovative approach teaching students about BIM and MEP in architecture, Digital Fabrication with second hand materials

Internship
Yilan, Taiwan Jul 2021 - Jul 2024
Global Reengineering Co., Ltd.
Responsibilities: Optimisation of prefabricated concrete façade panels, structural skeleton visualisation, and design coordination.

Project Assistant
Taipei, Taiwan, Sep 2020 - Jul 2022
Research Team: GAS Lab, Prof ShenGuan Shih

Skill

Technical: Various Softwares (Rhino, Revit, 3D Max, AutoCad, Sketchup, Blender etc.), Programming (C#, C++, Javascript, Python, Typescript, R, etc), Git, Json, tool (Respberry Pi), CNC Fabrication

Scientific working and Writing : LaTeX, Microsoft Office,

AI and Computational tool: Stable Diffusion, Chat GPT,

About

mike861104@gmail.com

Tsung-Wei-Cheng

宗偉-鄭-b76a45251

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Lectures and Talks

Procedural Design for Interwoven Structures using SL Blocks: A computational Framework for Modular Architecture
ICCEA 2025 - Best Presentation Prize

Game-based Learning with Minecraft: Fostering Engagement in Mechanical-Electrical-Plumbing Systems Education
Asia-Pacific Conference on Education, Teaching & Technology 2025

Reducing Panel Complexity in Topological Interlocking Assemblies on Curved Surfaces
CADDRIA Conference 20th 2025

Lecturer at NUK
Teaching with Prof. YuPin Ma, National University of Kaohsiung
Nov 2023 - JUN 2025
Digital Fabrication
Minecraft Serious Gaming
Lego BIM Workshop
Rhino Advance Lecture

Design& Assembly: Symmetry Informed Assembly of Osteomorphic Block in Computer-Aided Design
The Research Exchange Day, Digital Design Unit at Technische Universität Darmstadt, Apr, 2024

Rule-Based Generation of Interwoven Assembles in Architectural Design
CADDRIA Conference 19 th 2024

Grasshopper Lecture
Teaching with Prof. Fufei Yong,
National Taiwan University of Technology and Science, Nov 2023

Publications (Selected)

Tsung-Wei Cheng, Felicia Wagiri, Kevin Harsono, Ching-Yen Cheng, Shen-Guan Shih. (2025) Procedural Design for Interwoven Structures using SL Blocks: A computational Framework for Modular Architecture

Yu-Pin Ma, Tsung-Wei Cheng*, Rou Syuan Guo, Kevin Harsono, Felicia Wagiri, Shen-Guan Shih (2025). Game-based Learning with Minecraft: Fostering Engagement in Mechanical-Electrical-Plumbing Systems Education

Kevin Harsono, Tsung-Wei Cheng*, Felicia Wagiri, Ye Yint Aung, Shen-Guan Shih (2025) Modular and mortarless masonry: design, fabrication, and reusability assessment framework for topological interlocking assembly

Tsung-Wei Cheng, Kevin Harsono, Yuxi Lui, Ching-Yen Chen, Shen-Guan Shih, Oliver Tessmann. (2024). Beyond Babel: Towering With Minimal Communication. *Proceedings of the IASS 2024 Symposium.* International Association for Shell and Spatial Structures (IASS 2024). Switzerland.

Felicia Wagiri, Shen-Guan Shih, Yu-Chuan Kao, Tsung-Wei Cheng, Mu-Kuan Lu. (2024). Bending-active molds for pre-fabricated concrete shells. *Proceedings of the IASS 2024 Symposium.* Redefining the Art of Structural Design (IASS 2024). Switzerland.

Yuxi Liu, Boris Belousov, Tim Schneidern, Kevin Harsono, Tsung-Wei Cheng, Shen-Guan Shih, Oliver Tessman (2024). Advancing Sustainable Construction: Discrete Modular Systems & Robotic Assembly. *Sustainability* 16(15):6678. https://doi.org/10.3390/su16156678

Kevin Harsono, Shen-Guan Shih, Ye Yint Aung, Felicia Wagiri, Tsung-Wei Cheng (2024). RULE-BASED GENERATION OF INTERWOVEN ASSEMBLIES IN ARCHITECTURAL DESIGN: A COMPUTATIONAL APPROACH INTEGRATING ATTRIBUTE GRAMMAR. *Proceedings of the 29th International Conference on Computer-Aided Architectural Design Research in Asia (CAADRIA 2024).* At: Singapore.

Cheng, T.W., Wu, B.R., Chiang, Y.T., Liao, C.T. (2018). The Study on Unlift of Exterior Tile in Nanzih District of Kaohsiung City Processings of the 30th Annual Conference of the Architecture Institute of Taiwan

Tsung-Wei Cheng. (2022), Production Rules for the Arrangement of Osteomorphic Block.

Education

Exchange Researcher
Darmstadt, Germany, April 2024
Auto-Replicating Robotic Assembly of SL Blocks for the Hierarchical and Reconfigurable Construction
Project headed by Prof. ShenGuan Shih, and Prof Oliver Tessmann

Exchange Student
Berlin, Germany, Sep 2022-Mar 2023
Circular Economy and Ontology of Building Information Modelling
Project headed by Prof. Timo Hartmann
the Department of Civil Engineering, Technische Universität Berlin

Master of Architecture
Taipei, Taiwan, Sep 2020-Jul 2023
Production Rule for the design patterns of modular design system
Thesis: Production Rules for the Arrangement of Osteomorphic Block
Supervision headed by Prof. ShenGuan Shih
GAS Lab, National Taiwan University of Technology and Science

Bachelor of Creative Design and Architecture
Taipei Taiwan, Sep 2016-Jul 2020
Colledge of Humanities and Social Sciences
National University of Kaohsiung

Funding

DAAD-PPP_Program (2nd)- Auto-Replicating Robotic Assembly for SL Blocks for the Hierarchical and Reconfigurable Construction
Jan 2025 - Present
National Science and Technoloygy Council

STEAM Course
Jan 2025 - Present
National University Kaohsiung

The Architecture of Shape Compiler
Mar 2025 - Present
National Science and Technology Council

DAAD-PPP_Program (1st)- Auto-Replicating Robotic Assembly for SL Blocks for the Hierarchical and Reconfigurable Construction
Jan 2024 - Dec 2025
National Science and Technoloygy Council

Digital Application for Sustainable Built Environment Program
Sep 2023 - Mar 2024
National Taiwan University of Technology and Science

Exploring the Use of Truchet Tiling in Architectural Facade Design
Sep 2022 - Mar 2023
Scholarship of Chung Hwa Rotary Education Foundation